# Resource Allocation in Multi-User Cellular Networks: A Transformer-Based Deep Reinforcement Learning Approach

**Zhao Di[1], Zheng Zhong[2], Qin Pengfei[2], Qin Hao[1,*], Song Bin[1]**

[1] State Key Laboratory of Integrated Services Networks, Xidian University, 710071, China
[2] China Academy of Space Technology, Beijing 100081, China
[*] The corresponding author, email: hqin@mail.xidian.edu.cn

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**Abstract:** To meet the communication services with diverse requirements, dynamic resource allocation has shown increasing importance. In this paper, we consider the multi-slot and multi-user resource allocation (MSMU-RA) in a downlink cellular scenario with the aim of maximizing system spectral efficiency while guaranteeing user fairness. We first model the MSMU-RA problem as a dual-sequence decision-making process, and then solve it by a novel Transformer-based deep reinforcement learning (TDRL) approach. Specifically, the proposed TDRL approach can be achieved based on two aspects: 1) To adapt to the dynamic wireless environment, the proximal policy optimization (PPO) algorithm is used to optimize the multi-slot RA strategy. 2) To avoid co-channel interference, the Transformer-based PPO algorithm is presented to obtain the optimal multi-user RA scheme by exploring the mapping between user sequence and resource sequence. Experimental results show that: i) the proposed approach outperforms both the traditional and DRL methods in spectral efficiency and user fairness, ii) the proposed algorithm is superior to DRL approaches in terms of convergence speed and generalization performance.

**Keywords:** dynamic resource allocation; multi-user cellular network; spectrum efficiency; user fairness

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## I. INTRODUCTION

The fifth generation mobile communication technology (5G) has become the reality and demand for 2020 and beyond. As a significant manifestation of 5G, the popularity of consumer-centric and data-intensive applications [1] has led to a boom of device density. A large number of terminal devices accessing the network generate greater demand for spectral bandwidth [2]. To meet the ever-increasing spectral demand, base stations (BSs) have been more densely deployed [3]. However, both the massive access of terminals and the dense deployment of BSs inevitably lead to severe inter-cell interference [4], which greatly reduces the spectrum efficiency (SE) of cell-edge users [5] in multi-user cellular networks. As a solution to improve the SE performance of both cell-edge users and cell-center users, resource allocation (RA) [6, 7] becomes a critical research focus.

Recently, the idea of incorporating data-driven approaches in wireless resource management has developed rapidly [8, 9]. In particular, reinforcement learning (RL) has been widely used to tackle RA problems to achieve near-optimal objectives [10]. In the RL-based methods, the RL agent is to maximize long-term returns, rather than simply getting the current optimal return. This is important for time-varying dynamic systems, especially in wireless networks [11]. However, next-generation wireless networks (B5G and 6G) must accommodate surging growth in mobile data traf-



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fic, and support an increasingly high density of mobile users [12], which greatly increases the complexity of RA problems. In this case, deep reinforcement learning (DRL) has been proposed to deal with the large-scale RA problem in wireless networks [13]. Although the DRL-based RA methods has been proven to achieve good performance [14], several problems remain to be addressed, such as the lack of theoretical connections and explanations [15], the constraint of a fixed number of requesting users [16] and the ultra-high computational complexity [17].

Inspired by the information processing mechanism of the human brain, we consider that attention is a promising solution to the above-mentioned problems. As an important component of neural architectures, attention can enhance the interpretability of the trial-and-error learning [18]. Transformer, proposed by the Google Brain in [19], is a model that completely relies on the attention mechanism. Transformer has been applied to the field of wireless communication [20, 21]. In order to address the above-mentioned problems in DRL-based RA methods, we consider that it is feasible to establish a Transformer-based DRL (TDRL) framework due to the following three motivations.

- *Establish theoretical connections.* The unreasonable resource allocation scheme [22] and the dense user location distribution [23] may cause serious co-channel interference, which leads to poor RA performance. By extracting the attention pattern in user location distribution and resource allocation scheme, the attention mechanism of Transformer may establish a theoretical connection between deep learning network and co-channel interference.
- *Remove constraints on number of requesting users.* Affected by the structure of deep learning networks, existing works [24] have designed a fix-dimension neural network to accommodate a fixed number of requesting users, which is not consistent with the real scenario. The mask setting of Transformer is able to convert fixed-length requesting users into variable-length ones by masking inactive users, which is more suitable for real scenarios.
- *Reduce computational complexity.* The two factors, large-scale user numbers and multi-dimension user information [25], may lead to an enormous state/action space, which greatly increase the complexity of large-scale RA problems. With the aid of Transformer, the large-scale RA can be performed by making decisions one by one. This recursive output generation may be an effective way to reduce the action space.

In this paper, we focus on the multi-slot and multi-user resource allocation (MSMU-RA) in a downlink cellular scenario. Compared with single-slot resource allocation, multi-slot resource allocation takes continuity into consideration, including the temporal variations in channel conditions and resource availability, thus, it is capable of avoiding interference to produce a long-term scheduling benefits. Considering the resource allocation over a period of time, the optimization goal tends to be diversified and the decision space tends to be supersized, which brings a hard challenge to the MSMU-RA problem. To deal with this challenge, we propose a TDRL framework composed of DRL in conjunction with Transformer. Specifically, we utilize DRL tools to describe the dynamic interactions between the RA agent and the environment, and propose a Transformer-based decision-making approach to study the co-channel interference mechanism. This is the first attempt at applying Transformer to the resource allocation problem. The major contributions of this paper are summarized as follows:

1) The multi-slot RA problem is modeled as a Markov decision process (MDP), and is solved by the proximal policy optimization (PPO) algorithm. The actor network of the PPO algorithm is responsible for generating RA actions, and the critic network of the PPO algorithm evaluates these actions to enhance convergence performance. Thus, the agent learns the optimal RA policy by taking the largest possible improvement step without performance collapse.
2) The multi-user RA problem is described as a mapping from user sequence to resource sequence, and is solved by a Transformer-based structure. To avoid co-channel interference, the encoder of Transformer is applied to characterize spatial correlations within the user sequence, and the decoder of Transformer is utilized to capture scheduled dependences within the resource sequence.
3) In performance comparison and testing, we build a wireless communication network simulation



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platform that connects the NS3 simulator [26] and OpenAI Gym [27] to achieve system-level simulation. Simulation results demonstrate that, our proposed algorithm outperforms traditional methods and other DRL-based approaches in terms of SE and user fairness.

The remainder of this paper is organized as follows. Section II briefly introduces related works. The system model is described in Section III, and the optimization problem is analyzed. Next, we propose the TDRL algorithm in Section IV. Then, simulation performance and discussion are presented in Section V. Finally, Section VI concludes the paper.

## II. RELATED WORKS

### 2.1 Traditional RA Methods

Resource allocation in wireless communication networks has been investigated for decades. The MSMU-RA problem in cellular networks is a non-convex, NP-hard and multi-objective optimization problem [28]. Existing researches can be classified into BS-level and user-level methods. The BS-level scheduling methods divide the cell coverage into a center area and an edge area, and allocate specific resources to a certain region. For example, soft frequency reuse (SFR) and fractional frequency reuse (FFR) [29] can fit with the traffic distribution by adjusting the power threshold ratio of the subcarrier to the main carrier. On the other side, the user-level scheduling methods are mostly optimized by heuristic algorithms [30] to obtain sub-optimal solutions. Heuristic algorithms include round robin (RR) scheduling algorithm [31], maximum of carrier-to-interference ratio (Max C/I) algorithm [32] and proportional fairness (PF) algorithm [33]. In comparison, it is difficult for the RR algorithm to perform well in the overall throughput of the network, and the Max C/I algorithm is an unfair scheme for cell-edge and deep-fading users. In order to achieve a trade-off between throughput and fairness, the PF algorithm is proposed to balance system performance and user experience. However, the above-mentioned traditional methods are unable to adapt to a dynamic network environment, making it impossible to find the optimal solution.

### 2.2 DRL-Based RA Methods

Driven by artificial intelligence (AI), existing works have applied a general framework known as DRL for resource allocation. An overview of DRL-based resource management methods was given in [34], and these methods allow network entities to learn and build knowledge about the wireless environments and thus make optimal decisions. However, the above methods that require complete knowledge of the communication systems are inefficient in scenarios with imperfect or outdated channel state information (CSI) [35]. In order to cope with this problem, the authors in [36] proposed a deep Q-networks (DQN)-based RA algorithm to maximize the sum-rate for any given limited information on instantaneous CSI. In addition to the value-based DRL methods mentioned above, there is also a large category of policy-based approaches. In [37], a dynamic downlink power control problem was considered, and the authors applied policy-based REINFORCE and deep deterministic policy gradient (DDPG) algorithms to maximize the sum-rate. Moreover, multi-agent DRL methods are studied to reduce the computational complexity of centralized DRL approaches. The authors in [38] presented a distributed optimization method based on multi-agent DRL in the downlink of heterogeneous cellular networks to achieve the maximum long-term overall network utility while guaranteeing the quality of service (QoS) requirements of users.

### 2.3 Transformer

Transformer has made significant contributions in the fields of image recognition [39], machine translation [40] and text recognition [41]. Moreover, Transformer has been applied to the field of wireless communication. The authors in [42] explored Transformer architectures for load and energy forecasting in smart cities, in which Spatial Transformer sub-modules are applied to capture hidden spatial representation and Temporal Transformer sub-modules are employed to model temporal dependencies within input sequences. In [43], the authors implemented a Transformer model to infer a placement policy of virtualization network functions (VNFs) aimed at minimizing the overall power consumption. As far as we know, the attention-based Transformer model is rarely noticed in the field of resource allocation.



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## III. SYSTEM MODEL AND PROBLEM FORMULATION

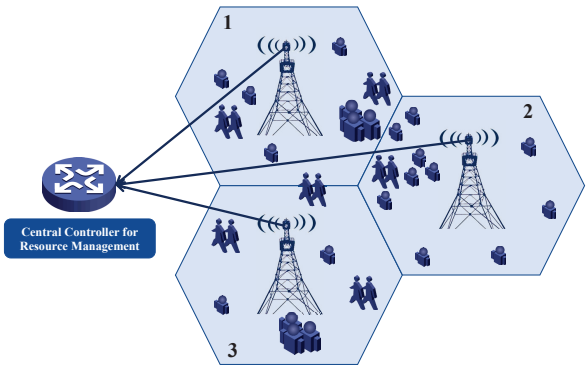



**Figure 1.** *System model.*

### 3.1 System Model

All studies on the MSMU-RA problem are based on the downlink of cellular networks due to the scarcity of downlink spectrum resources. The multi-user cellular network is shown in Figure 1. It consists of several cells, and one BS is deployed in the center of each cell. Thus, the BSs in the system can be denoted by $\mathcal{B} = \{1, 2, \ldots, B\}$. Assuming that a certain number of users are simultaneously served by the BS, all users in the system can be represented by $\mathcal{U} = \{1, 2, \ldots, U\}$. In addition, we assume that a central controller is set up to make RA decisions for the overall system.

Resource allocation in a cellular network is essentially a process of scheduling available resource blocks (RBs) to requesting users. We regard a transmission time interval (TTI) as a unit to perform resource allocation. In TTI $t$, the status of all users can be expressed as $\mathcal{X}_t^{\mathcal{U}} = \left\{x_t^1, x_t^2, \ldots, x_t^U\right\}$, where $x_t^u \in \{0, 1\}$. $x_t^u = 1$ means that user $u$ makes a request in TTI $t$, and $x_t^u = 0$ means that the user is inactive. We assume that users in the same cell are served on orthogonal RBs to avoid intra-cell interference, and the set of orthogonal RBs is denoted as $\hat{\mathcal{C}} = \{1, 2, \ldots, C\}$. Besides, we set the frequency reuse factor to 1, which means that all orthogonal RBs used in a cell can be reused in all neighboring cells. Thus, each RB in the system is represented as $\mathcal{C} = \left\{(b, c) \,|b \in \mathcal{B}, c \in \hat{\mathcal{C}}\right\}$ where $b$ is the BS that provides RB $c$. In TTI $t$, the status of all RBs can be written as $\mathcal{X}_t^{\mathcal{C}} = \left\{x_t^{1,1}, x_t^{1,2}, \ldots, x_t^{B,C}\right\}$ where $x_t^{b,c} = 0$ means that RB $c$ of BS $b$ is free and $x_t^{b,c} = 1$ means this RB is unavailable. Note that not all RBs are available in each TTI. After allocation, the RA results in TTI $t$ can be expressed as $\mathcal{W}_t = \left\{w_t^{1,1}, w_t^{1,2}, \ldots, w_t^{B,C}\right\}$ where $w_t^{b,c} \in \mathcal{U} \cup \{0\}$. $w_t^{b,c} = u$ indicates that user $u \in \mathcal{U}$ served by BS $b$ occupies RB $c$ of the same BS, and $w_t^{b,c} = 0$ indicates that this RB is unallocated.

In this paper, the channel gain between BS $b$ and user $u$ in TTI $t$ is denoted by $h_u^b[t]$. The modeling of $h_u^b[t]$ mainly takes the path loss into consideration. We calculate the channel gain in dB based on the log-distance propagation loss model [44] with three distance fields. The channel gain $h_u^b[t]$ is written by

$$h_u^b[t] = L_0 + \sum_{i=0}^{j} 10 \cdot n_i \log_{10}\left(\frac{d_{i+1}}{d_i}\right), 0 \le j \le 2, \tag{1}$$

where $L_0$ is the path loss at reference distance. $d_i\,(0 \le i \le 2)$ are three distance fields. Accordingly, $n_i\,(0 \le i \le 2)$ are the path loss distance exponents.

Assuming that the interference only comes from neighboring cells, the received signal to interference plus noise ratio (SINR) over RB $c$ of BS $b$ in TTI $t$ can be expressed as

$$\xi_c^b[t] = \frac{S_c^b[t]}{I_c^b[t] + \sigma^2}, \tag{2}$$

where we denote

$$S_c^b[t] = \left|h_{w_t^{b,c}}^b[t]\right|^2 \cdot p_c^b[t], \tag{3a}$$

$$I_c^b[t] = \sum\nolimits_{n=1, n\neq b}^{B} \left|h_{w_t^{n,c}}^n[t]\right|^2 \cdot p_c^n[t]. \tag{3b}$$

$p_c^b[t]$ and $p_c^n[t]$ represent the transmit powers on RB $c$ of BS $b$ and BS $n$ in TTI $t$, respectively. $\sigma^2$ is the average power of additive white Gaussian noise (AWGN).

### 3.2 Problem Formulation

In this paper, the studied MSMU-RA problem is formulated as

$$\begin{aligned} \max_{\mathcal{W}=(\mathcal{W}_1, \mathcal{W}_2, \ldots, \mathcal{W}_T)} \quad & \mathbb{E}\left[\sum\nolimits_{t=1}^{T}\left(\alpha_1 * \frac{\Psi_t}{\Psi_{max}} + \alpha_2 * \Gamma_t\right)\right] \\ s.t. \quad & w_t^{b,c} = 0, \quad \text{if } x_t^{b,c} = 1. \end{aligned} \tag{4}$$


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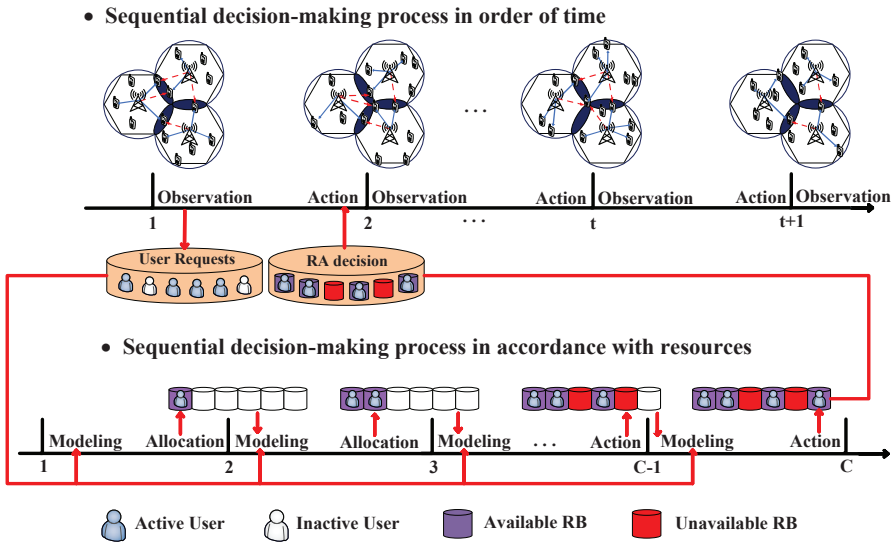



**Figure 2.** *The dual-sequence decision-making process.*

Here, $\mathcal{W} = (\mathcal{W}_1, \mathcal{W}_2, ..., \mathcal{W}_T)$ is the joint RA vector on $[1, T]$. It can be seen from Eq. (4) that we aim to maximize the system SE while avoiding unfair allocation with limited available spectrum resources. On the one side, $\Psi_t$ is the SE of the system in TTI $t$ and is expressed by Shannon's formula as

$$\Psi_t = \sum_{b=1}^{B} \sum_{c=1}^{C} \log_2 \left(1 + \xi_c^b \left[t\right]\right), \tag{5}$$

and $\Psi_{max}$ is the theoretical boundary of the system SE calculated by the maximum transmission capacity of each RB. Thus, the value of $\Psi_t / \Psi_{max}$ is normalized in $[0, 1]$. On the other side, $\Gamma_t$ represents the fairness of user capacity over $[1, t]$ and is designed based on Jain's Fairness Index [45] as

$$\Gamma_t = \frac{\left(\sum_{u=1}^{U} \left(\sum_{k=1}^{t} \Psi_k^u\right)\right)^2}{U \cdot \sum_{u=1}^{U} \left(\sum_{k=1}^{t} \Psi_k^u\right)^2}, \tag{6}$$

which is also a value of $[0, 1]$. $\Psi_k^u$ is the SE of user $u$ and is defined by

$$\Psi_t^u = 1\left(w_t^{b,c} = u\right) \cdot \sum_{b=1}^{B} \sum_{c=1}^{C} \log_2 \left(1 + \xi_c^b \left[t\right]\right), \tag{7}$$

where $1\left(w_t^{b,c} = u\right)$ is a logical function to determine whether RB $c$ of BS $b$ is allocated to user $u$. Mathematically, the logical function is written as

$$1\left(w_t^{b,c} = u\right) = \begin{cases} 1, & \text{if } w_t^{b,c} = u, \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

In addition, $\alpha_1$ and $\alpha_2$ are the weights for SE and user fairness performance, respectively.

The optimization problem in Eq. (4) is analyzed as follows. Firstly, we analyze the multi-user RA problem focusing on one TTI. The optimization objective is denoted as $\mathcal{J}\left[\mathcal{W}_t\right]$, and the mean of $\mathcal{J}\left[\mathcal{W}_t\right]$ can be expressed as

$$\begin{aligned} & \mathbb{E}_{\mathcal{W}_t = \left(w_t^{1,1}, w_t^{1,2}, \ldots, w_t^{B,C}\right)} \mathcal{J}\left[\mathcal{W}_t\right] \\ = & \sum_{m=1}^{M} p^m \left(w_t^{1,1}, w_t^{1,2}, \ldots, w_t^{B,C}\right) \cdot \mathcal{J}\left[\mathcal{W}_t^m\right] \\ = & \sum_{m=1}^{M} \prod_{b,c=1,1}^{B,C} p^m \left(w_t^{b,c} | w_t^{1,1:b,c-1}\right) \cdot \mathcal{J}\left[\mathcal{W}_t^m\right]. \end{aligned} \tag{9}$$

Here, $M$ is the total number of possible values of $\mathcal{W}_t$, and the maximum of $M$ is $(U+1)^{BC}$. It can be seen from Eq. (9) that, the global probability of a group of RBs for allocation can be decomposed into the product of the conditional probabilities of individual RB. Moreover, these conditional probabilities indicate that the allocated RBs have an impact on the next unallocated one. From the perspective of resources, we consider that the multi-user RA problem within a TTI is a sequential decision-making process.

However, Eq. (9) uses the mean of discrete random



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variables to estimate the value of $\mathcal{J}[\mathcal{W}_t]$, which inevitably leads to higher computational costs in large-scale scenarios. To overcome it, we parameterize the conditional probabilities in the following.

**Next, we analyze the multi-user RA problem over multiple TTIs.** We describe the RA policy on $[1, T]$ as a function with parameters $\theta$ as

$$\mathcal{W}^* = \arg\max_\theta \mathbb{E}_{\mathcal{W}\sim\pi_\theta(\mathcal{W})}\left[\sum_{t=1}^{T}\mathcal{J}[\mathcal{W}_t]\right], \quad (10)$$

aiming to make the optimal RA decisions based on the optimal policy by optimizing parameters $\theta$.

Thus, the objective function of the multi-TTI RA problem in Eq. (4) is further written as

$$\begin{aligned} &\mathbb{E}_{\mathcal{W}\sim\pi_\theta(\mathcal{W})}\left[\sum_{t=1}^{T}\mathcal{J}[\mathcal{W}_t]\right] \\ =\ &\int_{\mathcal{W}\sim\pi_\theta(\mathcal{W})} p_{\pi_\theta}(\mathcal{W})\cdot\left(\sum_{t=1}^{T}\mathcal{J}[\mathcal{W}_t]\right)\mathrm{d}\mathcal{W}. \end{aligned} \quad (11)$$

To maximize the optimization objective, we need to take the derivative of the objective function, and update the parameters in gradient direction. The derivative of Eq. (11) with respect to $\theta$ is expressed by

$$\begin{aligned} &\nabla_\theta\mathbb{E}_{\mathcal{W}\sim\pi_\theta(\mathcal{W})}\left[\sum_{t=1}^{T}\mathcal{J}[\mathcal{W}_t]\right] \\ =\ &\int_{\mathcal{W}\sim\pi_\theta(\mathcal{W})}\nabla_\theta p_{\pi_\theta}(\mathcal{W})\cdot\left(\sum_{t=1}^{T}\mathcal{J}[\mathcal{W}_t]\right)\mathrm{d}\mathcal{W} \\ =\ &\int_{\mathcal{W}\sim\pi_\theta(\mathcal{W})} p_{\pi_\theta}(\mathcal{W})\cdot\nabla_\theta\log p_{\pi_\theta}(\mathcal{W})\cdot\left(\sum_{t=1}^{T}\mathcal{J}[\mathcal{W}_t]\right)\mathrm{d}\mathcal{W} \\ =\ &\mathbb{E}_{\mathcal{W}\sim\pi_\theta(\mathcal{W})}\left[\nabla_\theta\log p_{\pi_\theta}(\mathcal{W})\cdot\left(\sum_{t=1}^{T}\mathcal{J}[\mathcal{W}_t]\right)\right] \\ =\ &\mathbb{E}_{\mathcal{W}\sim\pi_\theta(\mathcal{W})}\left[\left(\sum_{t=1}^{T}\nabla_\theta\log p_{\pi_\theta}(\mathcal{W}_t|s_t)\right)\cdot\left(\sum_{t=1}^{T}\mathcal{J}[\mathcal{W}_t]\right)\right], \end{aligned} \quad (12)$$

where the policy function $p_{\pi_\theta}(\mathcal{W})$ is defined as

$$p_{\pi_\theta}(\mathcal{W}) = p(s_1)\prod_{t=1}^{T} p_{\pi_\theta}(\mathcal{W}_t|s_t)\, p(s_{t+1}|s_t, \mathcal{W}_t). \quad (13)$$

The policy function is responsible for executing RA decisions based on the observed information. The conditional probability $p_{\pi_\theta}(\mathcal{W}_t|s_t)$ in Eq. (12) means that the RA decision taken in each TTI depends on the current observation, and varies in time. Furthermore, RA decisions $\mathcal{W}_t$ are arranged in chronological order to obtain the sequential RA strategies over $[1, T]$.

In summary, the MSMU-RA problem is a dual-sequence decision-making process as shown in Figure 2. More specifically, the two sequential decision-making models are:

i) The RA problem in one TTI is a sequential decision-making model in accordance with resources;

ii) The RA problem over multiple TTIs is a sequential decision-making model in order of time.

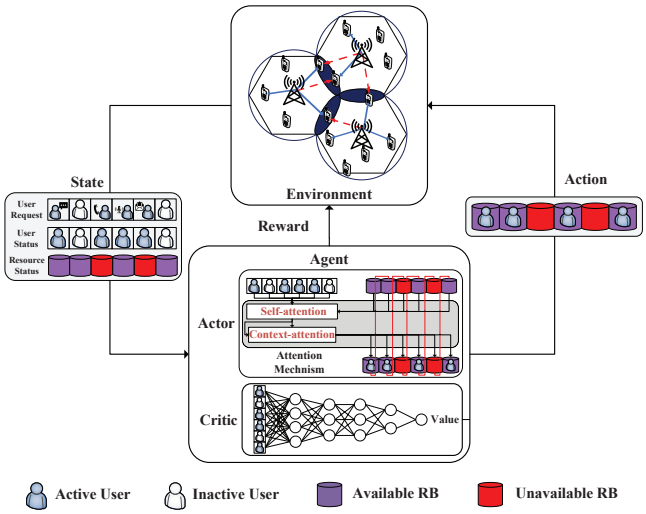



**Figure 3.** *The TDRL algorithm.*

## IV. THE TRANSFORMER-BASED DEEP REINFORCEMENT LEARNING ALGORITHM

This section introduces the TDRL algorithm in multi-user cellular networks, as shown in Figure 3. To be specific, we utilize RL tools to model the dynamic interactions between the RA agent and the environment, and adjust the RA strategy based on "observation-actor-critic-returns". With the aim of avoiding co-channel interference, we design a Transformer-based actor framework for decision-making, and employ the attention mechanism to explore spatial correlations and scheduled dependencies.



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### 4.1 Preliminaries

An RL approach is usually to solve an MDP. An MDP is generally depicted by a five-tuple model, which is $\langle \mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{P}, \gamma \rangle$. Specifically, $\mathcal{S}$, $\mathcal{A}$, $\mathcal{R}$, $\mathcal{P}$ and $\gamma$ represent the state, action, reward, transition probability and discount factor, respectively. At time $t$, the agent observes the environment for state $s_t \in \mathcal{S}$, and then selects action $a_t \in \mathcal{A}$ in accordance with the policy $\pi(a_t|s_t)$. After the action is performed, the environment is transitioned to a new state $s_{t+1} \in \mathcal{S}$ with a probability of $p(s_{t+1}|s_t, a_t)$. Meanwhile, the agent will get an immediate reward $r_t \in \mathcal{R}$. The goal of RL is to find the optimal policy $\pi^*(a_t|s_t)$ by maximizing the long-term cumulative returns $R_t = \sum_{t=0}^{\infty} \gamma^t r_{t+1}$, that is $\pi^* = \arg\max_\pi \mathbb{E}_\pi[R_t]$. Specially, $R_t = \sum_{t=0}^{T} \gamma^t r_{t+1}$ is called the episode returns if the MDP has an explicit termination state.

In this paper, a neural network called "actor" is employed to approximate the policy function with parameters $\theta^A$. By the stochastic gradient descent (SGD) method, the parameters $\theta^A$ are updated to gradually improve the RA policy. Moreover, the update size of $\theta^A$ can be determined by the cumulative returns $R_t$. Instead of using the environmental feedback $R_t$, we adopt a neural network with parameters $\theta^C$ called "critic" to estimate the values. In this way, the paradigm of "observation-actor-critic-returns" is built to describe the dynamic interactions between the RA agent and the cellular network. However, the determination of the update size remains a problem. To solve the problem, the proximal policy optimization (PPO) algorithm [46] takes the largest possible improvement step on a policy using the data we currently have, without stepping too far that may accidentally cause performance collapse. The objective function of the actor network is given by

$$\mathcal{J}\left(\theta^A\right) = \frac{1}{N}\sum_{t=1}^{N} \min \left[ \begin{array}{l} \rho_t\left(\theta^A\right) A\left(s_t, a_t; \pi_{\theta^A_{\text{old}}}\right) \\ \text{clip}\left(\cdot\right) A\left(s_t, a_t; \pi_{\theta^A_{\text{old}}}\right) \end{array} \right]. \tag{14}$$

Here, $A(s_t, a_t) = Q(s_t, a_t) - V(s_t)$ represents the advantage function, and $\rho_t\left(\theta^A\right) = \frac{\pi\left(a_t|s_t;\theta^A\right)}{\pi\left(a_t|s_t;\theta^A_{\text{old}}\right)}$ is the ratio of probabilities of new and old policies. In order to ensure the optimization stability, the PPO algorithm delimits the trust region, and the boundary is given by the clip function $\text{clip}(\cdot) = \text{clip}\left(\rho_t\left(\theta^A\right), 1-\epsilon, 1+\epsilon\right)$. In addition, the objective function of the critic network is given by

$$\mathcal{J}\left(\theta^C\right) = -\sum_{t=1}^{T}\left(\sum_{t'>t}\gamma^{t'-t} r_{t'} - V\left(s_t; \pi_{\theta^C}\right)\right)^2. \tag{15}$$

**Table 1.** *Request parameters.*

| Parameter | Physical Meaning |
|---|---|
| $\text{IDX}_t^q$ | Request ID (Session ID) |
| $\text{Cell_ID}_t^q$ | Serving cell ID of the requesting user |
| $\text{UE_ID}_t^q$ | Requesting user ID |
| $\text{D}_t^q$ | Distance between the BS and the user |
| $\text{TD}_t^q$ | Transmission Delay |
| $\text{QL}_t^q$ | Queue length |
| $\text{ArrBytes}_t^q$ | Number of newly arrived bytes |
| $\text{TxedBytes}_t^q$ | Number of last transmitted bytes |
| $\overline{\Psi}_t^q$ | Average throughput |
| $\text{Req_RB}_t^q$ | Number of RBs required |
| $\text{WB_CQI}_t^q$ | Wide-band CQI |
| $\text{SB_CQI}_t^q$ | Sub-band CQI |

### 4.2 DRL Design

**State:** The information of the cellular network perceived by the RA agent is denoted by

$$s_t = \left\{\mathcal{X}_t^{\mathcal{U}}, \mathcal{X}_t^{\mathcal{C}}, \mathcal{X}_t^{\mathcal{Q}}\right\}, s_t \in \mathcal{S}, \tag{16}$$

where $\mathcal{X}_t^{\mathcal{U}}$ and $\mathcal{X}_t^{\mathcal{C}}$ represent the status of users and RBs, respectively. $\mathcal{X}_t^{\mathcal{Q}}$ is the set of user requests.

Consistent with Section III, $\mathcal{X}_t^{\mathcal{U}}$ is represented as

$$\mathcal{X}_t^{\mathcal{U}} = \left\{x_t^1, x_t^2, ..., x_t^U\right\}, x_t^u \in \{0, 1\}, \tag{17}$$

which reflects the state of each user (active/inactive) in each TTI. The RA agent collects $\mathcal{X}_t^{\mathcal{U}}$ to set an allocation mask, which ensures that inactive users are not selected for scheduling.

Also, $\mathcal{X}_t^{\mathcal{C}}$ is written as

$$\mathcal{X}_t^{\mathcal{C}} = \left\{x_t^{1,1}, x_t^{1,2}, ..., x_t^{B,C}\right\}, x_t^{b,c} \in \{0, 1\}, \tag{18}$$

which describes the initial state of each RB (available/unavailable) in each TTI.

In addition, $\mathcal{X}_t^{\mathcal{Q}}$ is defined as

$$\mathcal{X}_t^{\mathcal{Q}} = \left\{x_t^q | q \in \left\{\mathcal{U} | x_t^u = 1, u \in \mathcal{U}\right\}\right\}, \tag{19}$$



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which is the set of request information of requesting users. $x_t^q$ is the information of request $q$ observed by the RA agent, and the parameters in $x_t^q$ are summarized in Table 1.

**Action:** For the MSMU-RA problem in a multi-user cellular system, the action space is discrete. That is, available RBs are assigned to requesting users, which is expressed as

$$a_t = \begin{bmatrix} w_t^{1,1} & w_t^{1,2} & \cdots & w_t^{1,C} \\ w_t^{2,1} & w_t^{2,2} & \cdots & w_t^{2,C} \\ \vdots & \vdots & \ddots & \vdots \\ w_t^{B,1} & w_t^{B,2} & \cdots & w_t^{B,C} \end{bmatrix}, a_t \in \mathcal{A}. \tag{20}$$

The element $w_t^{b,c}$ represents the user selected by RB $c$ of BS $b$ in TTI $t$, which is generated based on the probability distribution

$$\left[P_0^{b,c}, P_1^{b,c}, ..., P_u^{b,c}, ..., P_U^{b,c}\right]. \tag{21}$$

Here, $P_0^{b,c}$ is the probability that RB $c$ of BS $b$ is not scheduled, and $P_u^{b,c}$ is the probability that RB $c$ of BS $b$ is allocated to user $u$. Note that the row sum in $a_t$ should always be equal to 1.

Additionally, a recursive action generation approach is adopted in this paper. We assume that the allocation order of all RBs is to allocate RBs in one BS first, and then allocate that of other BSs. This order facilitates the central control to avoid co-channel interference, because the subsequent allocation is on the basis of previous RA results. To be specific, a start flag is set and fed into the policy network. In the first loop, we predict the allocation result $w_t^{1,1}$ based on the output vector $\left[P_0^{1,1}, P_1^{1,1}, ..., P_u^{1,1}, ..., P_U^{1,1}\right]$. Then, $w_t^{1,1}$, along with the start flag are used as input to predict the next allocation $w_t^{1,2}$ with the probability $p\left(w_t^{1,2}|w_t^{1,1}\right)$. The loop is repeated until the allocation of all available RBs in TTI $t$ is obtained.

**Reward:** Generally, the objective function is directly used as the immediate reward, so that the optimization goal can be quantified. In this paper, the reward is written as

$$r_t = \left(\alpha_1 * \frac{\Psi_t}{\Psi_{max}} + \alpha_2 * \Gamma_t\right), r_t \in \mathcal{R}. \tag{22}$$

Furthermore, we decompose $\Gamma_t$ from the perspectives of user satisfaction and resource utilization, thus $\Gamma_t$ is designed as

$$\begin{aligned} \Gamma_t = &\frac{1}{2} \cdot \frac{\left(\sum_{u=1}^{U}\left(\sum_{k=1}^{t}\varphi_k^u\right)\right)^2}{U \cdot \sum_{u=1}^{U}\left(\sum_{k=1}^{t}\varphi_k^u\right)^2} \\ &+ \frac{1}{2} \cdot \frac{\left(\sum_{b,c=1,1}^{B,C}\left(\sum_{k=1}^{t}\phi_k^{b,c}\right)\right)^2}{B \cdot C \cdot \sum_{b,c=1,1}^{B,C}\left(\sum_{k=1}^{t}\phi_k^{b,c}\right)^2}, \end{aligned} \tag{23}$$

where $\varphi_t^u$ characterizes the satisfaction of user $u$ in TTI $t$ and is represented as

$$\varphi_t^u = \frac{\mathrm{Alo}_t^u}{\mathrm{Req}_t^u}. \tag{24}$$

$\varphi_t^u$ is the ratio of the number of allocated RBs to requested RBs. Besides, $\phi_t^{b,c}$ describes the utilization over RB $c$ of BS $b$ in TTI $t$ and is denoted as

$$\phi_t^{b,c} = \frac{\mathrm{Ack}_t^{b,c}}{\mathrm{Cap}_t^{b,c}}, \tag{25}$$

which is the ratio of the number of acknowledge bytes to the maximum transmission capacity.

### 4.3 Attention-Based Transformer

As shown in Figure 4, the policy network is build based on the Transformer architecture. Transformer is a learning model that completely relies on the attention mechanism to characterize the global dependencies between inputs and outputs. The Transformer-based RA method is introduced as follows.

Firstly, the RA agent perceives the cellular network environment and collects the state information $\left\{\mathcal{X}_t^{\mathcal{U}}, \mathcal{X}_t^{\mathcal{C}}, \mathcal{X}_t^{\mathcal{Q}}\right\}$. Note that the dimensions of these elements, especially $\mathcal{X}_t^{\mathcal{Q}}$, expand severely with the number of users. As a solution to the curse of dimensionality, a technique, called user profiling [47], is adopted. The user profile is constructed by attaching "labels" to users, and these labels are highly refined feature identifications acquired by analyzing user information. Hence, we select features such as the BS, distance, queue length, channel quality and transmission delay to extract user labels. In this way, the state information with ultra-high dimensions is converted into a finite



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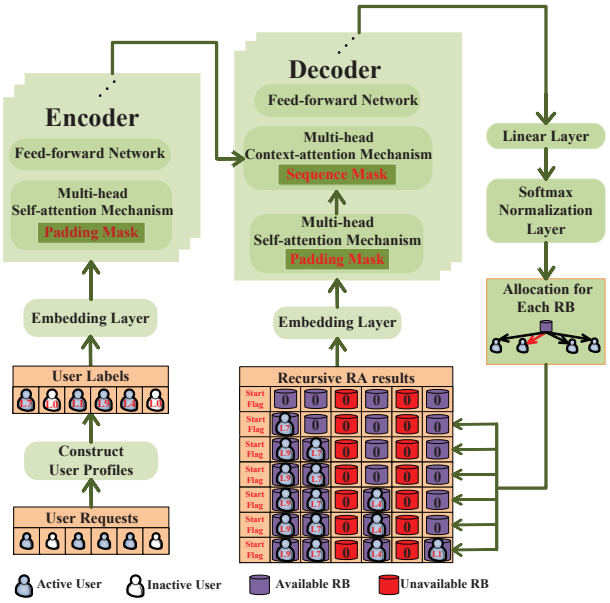



**Figure 4.** *The Transformer-based resource allocation.*

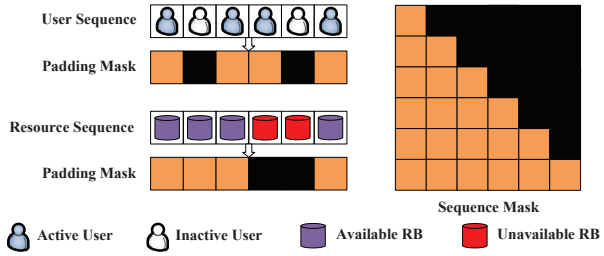



**Figure 5.** *Padding mask and sequence mask.*

set of user labels. At the same time, this conversion greatly reduces the state space.

Then, the embedding vectors about user labels are obtained through the embedding layer and inputted into the encoder of the Transformer architecture. The encoder is composed of $N$ identical layers, and each layer includes two modules: multi-head self-attention and feedforward network. For the multi-head self-attention mechanism, dealing with inactive users is tricky. Although the length of the input sequences for each batch is the same, the number of requesting users is different. It is necessary to fill in zeros for inactive users. These filled zeros are actually meaningless, so our attention should not focus on these positions. Thus, we use a padding mask to cover up the filled zeros, as shown in Figure 5. Specifically, we add negative infinity to the zero-filled places, so that the probability of these positions is close to 0 through the Softmax function. In this way, inactive users will

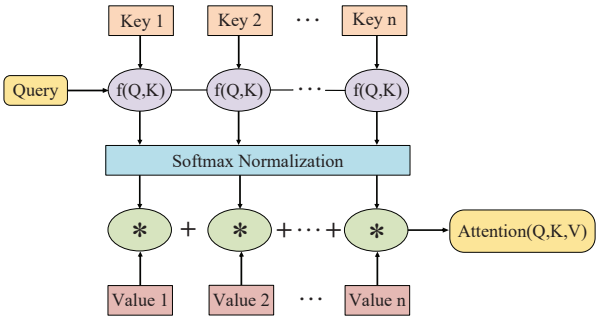



**Figure 6.** *The scaled dot-product attention mechanism.*

not have any effect during the parameter update process. Apart from that, the feedforward network is a fully connected network, which is designed as

$$FFN(x) = \max(0, xW_1 + b_1)W_2 + b_2. \quad (26)$$

Here, $W_1$ and $W_2$ are the weight matrices. Besides, $b_1$ and $b_2$ are the biases.

The decoder also consists of $N$ identical layers. Compared with the encoder, each layer of decoder has one more multi-head context-attention. To be consistent with Section 4.2, the decoder is designed to be recursive, which uses historical information as an additional input to predict the allocation. That is to say, the output of decoder over RB $c$ of BS $b$ should only depend on the RA results that ranked before it. Thereupon, we adopt a sequence mask to cover the information (filled zeros) after RB $c$ of BS $b$. The sequence mask is an upper triangular matrix, as shown in Figure 5. In the following, the self-attention vectors and encoding vectors of the decoder, along with the user sequence vectors, are fed into the multi-head context-attention layer. After a linear layer and a Softmax normalized layer, the probability distribution of resource allocation is obtained.

As mentioned above, the correlations and dependencies in MSMU-RA problems are characterized by self-attention and context-attention. The scaled dot-product attention mechanism is used in this paper. As shown in Figure 6, an attention function can be described as a query and a set of key-value pairs to an output. The scaled dot-product attention is given by

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (27)$$



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where $Q$, $K$, $V$ represent the query, key and value, respectively. $Q$, $K$ and $V$ are the same in self-attention. For the encoder, $Q$, $K$ and $V$ are the masked user vector to characterize spatial correlations within the user sequence, and for the decoder, $Q$, $K$ and $V$ are the masked resource vector to characterize scheduled dependences within the resource sequence. In context-attention, $K$ and $V$ are the output vector of the encoder, and $Q$ is the masked resource vector. The RB to be allocated is used as a query, and the RB is calculated with each user vector to obtain scheduled dependences. Then, combined with spatial correlations within the user sequence, the final allocation of the RB is determined. In addition, $1/\sqrt{d_k}$ is called the scaling factor, and $d_k$ is the dimension of the key. Instead of performing a single attention function, the multi-head attention is employed to linearly project the queries, keys and values $h$ times with different linear projections, which is represented by

$$\begin{aligned}&\text{MultiHead}\left(Q,K,V\right)\\=&\text{Concat}\left(\text{head}_1,\cdots,\text{head}_h\right)W^O,\\&\text{head}_i=\text{Attention}\left(QW_i^Q,KW_i^K,VW_i^V\right).\end{aligned}\tag{28}$$

Here, $W_i^Q \in \mathbb{R}^{d_{\text{model}}\times d_k}$, $W_i^K \in \mathbb{R}^{d_{\text{model}}\times d_k}$, $W_i^V \in \mathbb{R}^{d_{\text{model}}\times d_v}$ and $W^O \in \mathbb{R}^{hd_v\times d_{\text{model}}}$ are parameter matrices and $d_{\text{model}}$ is the dimension of the Transformer model. The parameters are not shared among multiple heads, which allows the model to learn relevant information in different sub-spaces.

### 4.4 TDRL Algorithm

The procedure of the Transformer-based PPO algorithm for the MSMU-RA problem in the multi-user cellular network is summarized in Algorithm 1. To ensure the real-time performance of resource allocation, we propose a offline-training and online-application pattern as follows.

**Algorithm 1.** *The Transformer-based PPO algorithm for the MSMU-RA problem in multi-user cellular networks.*

**Input:** request information of all users
**Output:** optimal allocation scheme for all RBs

**OFFLINE**
**if** initialization **then**
  initialize the actor network and parameters $\theta^A$
  initialize the critic network and parameters $\theta^C$
  reset gradients: $\mathrm{d}\theta^A \leftarrow 0$ and $\mathrm{d}\theta^C \leftarrow 0$
**else if** learning **then**
  **for** $i = 1$ to $N$ **do**
    **for** $t = 1$ to $T$ **do**
      observe user requests as state $s_t$
      actor generates action $a_t \sim \pi\left(a_t|s_t;\theta^A\right)$
      critic estimates value $V\left(s_t;\theta^C\right)$
      calculate return $r_t$ based on Eq. (22)
      estimate advantage $A\left(s_t,a_t;\pi_{\theta^A}\right)$
    **end for**
    collect data $\left\{s_t,a_t,r_t,V_(s_t),A(s_t,a_t)\right\}$
  **end for**
  **for** $j = 1$ to $M$ **do**
    calculate $\mathcal{J}\left(\theta^A\right)$ based on Eq. (14)
    **if** $\text{KL}\left[\pi_{\text{old}}^A|\pi_{\theta^A}\right] \le \text{KL}_{\text{target}}$ **then**
      update $\theta^A$ by a gradient method
      update policy: $\pi_{\theta^A_{\text{old}}} \leftarrow \pi_{\theta^A}$
    **end if**
  **end for**
  **for** $j = 1$ to $M$ **do**
    calculate $\mathcal{J}\left(\theta^C\right)$ based on Eq. (15)
    update $\theta^C$ by a gradient method
  **end for**
**end if**
**ONLINE**
collect information about users and resources
make the optimal RA scheme
users communicate on allocated resources

• **Offline Training:** The RA agent follows the "observation-actor-critic-returns" framework to interact with the multi-user cellular environment, so as to learn the optimal policy for maximizing the system SE while avoiding unfair allocation. Firstly, the agent collects the request information of all cells in a centralized way. Then, user requests are converted into user labels through user profiling technique. The sequence of user labels is fed into the Transformer-based actor network to generate the probability distribution of RA decisions. Afterwards, the actor network selects an action based on the current RA policy, and the critic network evaluates the advantage of the selected action. After performing the selected action, the agent obtains the returns closely related to the RA results. Finally, the learning of RA policy is designed in line with the PPO algorithm. Specifically, the network parameters of the actor-critic network are initialized before interacting with the environment. After that, the



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Transformer-based actor network updates parameters by a gradient method with respect to Eq. (14) within a KL restriction. Meanwhile, the critic network optimizes parameters according to Eq. (15). After a period of training, the agent has adapted to the complex and unstable channel conditions of the multi-user cellular communication system, and learns the optimal RA policy $\pi^*(a_t|s_t)$ to achieve a win-win situation between SE and fairness.

• **Online Application:** To ensure an unbiased update of the policy, the PPO algorithm takes the largest possible improvement step using the data we currently have, without stepping too far that may accidentally cause performance collapse. The optimal policy $\pi^*(a_t|s_t)$ obtained by the algorithm convergence is capable of ensuring the real-time resource allocation performance, because the communication environment has been fully explored in the training process. Thus, the real-time resource allocation can be well performed by loading the model parameters $\theta^*$ reserved in the training process. In addition, unlike the data fitting of supervised learning, RL aims to make optimal decisions in various environmental states through a reward and punishment mechanism. Therefore, RL has better adaptability and generalization in real-time resource allocation.

## V. SIMULATION AND EVALUATION

### 5.1 Simulation Settings

First of all, a system-level simulation is realized in this paper. We build a wireless communication environment simulation platform that connects the NS3 network simulator and OpenAI Gym. In this platform, the construction of the multi-cell and multi-user communication network is operated by the NS3 network simulator, which is responsible for the allocation of wireless resources and the mapping between logical channels and transmission channels at the media access control (MAC) layer. Besides, Gym, an API library developed by OpenAI, provides a unified interface for various environments and allows researchers to design RL algorithms. The experiments presented below are conducted in an Ubuntu operating system (CPU Intel core i7-10700K 3.8GHz; memory 16GB; GPU Intel Corporation Device 9bc5 (rev 05) and NVIDIA Corporation Device 2204 (rev a1)).

**Table 2.** *Simulation parameters.*

| Parameter | Value |
|---|---|
| Number of cells | 3 |
| Cell Coverage Radius | 800 m |
| System Bandwidth | 1.4 MHz |
| Terminal Mobility Model | Constant Position Mobility Model |
| Path Loss Model | Log-distance Propagation Loss |
| Path loss distance exponents | 1.9, 3.8, 3.8 |
| Three distance fields | 1 m, 200 m, 500 m |
| Number of users per cell | 16 |
| Number of RBs per cell | 6 |
| Weights of SE and fairness | 0.5, 0.5 |
| Capacity per RB | 100 Byte [48] |
| Tx Power of the BS | 30 dBm |
| Tx Power of the user | 20 dBm |
| Noise Figure | 5 dB |
| Traffic Model | ON/OFF Model |
| Mean of the ON process | 0.04 s |
| Mean of the OFF process | 0.004 s |
| Data rate in ON state | 500 kb/s |

**Table 3.** *Learning parameters.*

| Parameter | Value |
|---|---|
| Epochs | 1000 |
| Number of episodes per epoch | 5 |
| Number of steps per episode | 100 |
| Learning rate of Actor | 0.0003 |
| Learning rate of Critic | 0.001 |
| Activation Function | Tanh |
| Optimizer | Adam |
| Clip ratio | 0.2 |
| Discount factor | 0.99 |
| Target KL | 0.015 |
| Number of iterations | 50 |
| Number of heads | 4 |
| Dropout | 0.1 |
| Hidden layer size | 256, 512, 1024, 512, 256 |

In the simulation, a three-cell cellular network with 16 users per cell is considered, as illustrated in Figure 1. We assume that the duration of user communication follows a negative exponential distribution with parameter $\mu$. Therefore, the process of user requests follows the Poisson process. In each cell, users are located uniformly and randomly within the range $[0, R_c]$, where $R_c$ is the cell coverage radius. In each TTI, the constant position mobility model is set for all users. Additionally, the radio resource is organized into a number of RBs with 180 kHz per RB. The path loss model refers to Eq. (1). Here, the reference distance $d_0$ defaults to 1 m, and reference loss defaults to Friis Propagation Loss Model [49] with 5.15 GHz and thus $L_0 = 46.67$ dB. The rest of the key simulation param-



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eters are shown in Table 2.

In this paper, an actor-critic network framework is built. The Transformer-based actor network is designed as the encoder-to-decoder structure, and both the encoder and decoder contain 4 identical layers in addition to input and output layers. The critic network is a fully connected network with 7 layers, 5 layers of which are hidden layers. The input dimensions of both networks are $|S| = (16*1+6*1+16*1)*3 = 114$, and the output dimensions of the actor and critic networks are $|A| = 3*6*(6+1) = 126$ and 1, respectively. Then, we adopt the Tanh function as the activation function, and the Adam optimizer is used to update network parameters. Moreover, the number of episodes per epoch is 5, and the step count per episode is 100. The way that multiple episodes are used for one update can guarantee the accuracy and speed of the gradient optimization. The remaining learning parameters are reported in Table 3.

## 5.2 TDRL Algorithm Performance

In the following experiments, we evaluate the learning performance of our proposed algorithm on the basis of four metrics, which are defined as

- **Average Episode Returns:** the average of the episode returns is written as

$$\mathbb{E}\left[R_T\right] = \frac{1}{N}\sum_{n=0}^{N}\sum_{t=0}^{T}\gamma^t r_{t+1}^n. \tag{29}$$

- **Average Values:** the average of the summed state values for an episode is represented by

$$\mathbb{E}\left[V_T\right] = \frac{1}{N}\sum_{t=0}^{NT}V(s_t). \tag{30}$$

- **Average AckBytes:** the average of the summed acknowledge bytes for an episode is expressed as

$$\mathbb{E}\left[\text{AckBytes}_T\right] = \frac{1}{N}\sum_{t=0}^{NT}\sum_{q=1}^{Q}\text{AckBytes}_t^q. \tag{31}$$

- **Average NakBytes:** the average of the summed negative acknowledge bytes for an episode is given by

$$\mathbb{E}\left[\text{NakBytes}_T\right] = \frac{1}{N}\sum_{t=0}^{NT}\sum_{q=1}^{Q}\text{NakBytes}_t^q. \tag{32}$$

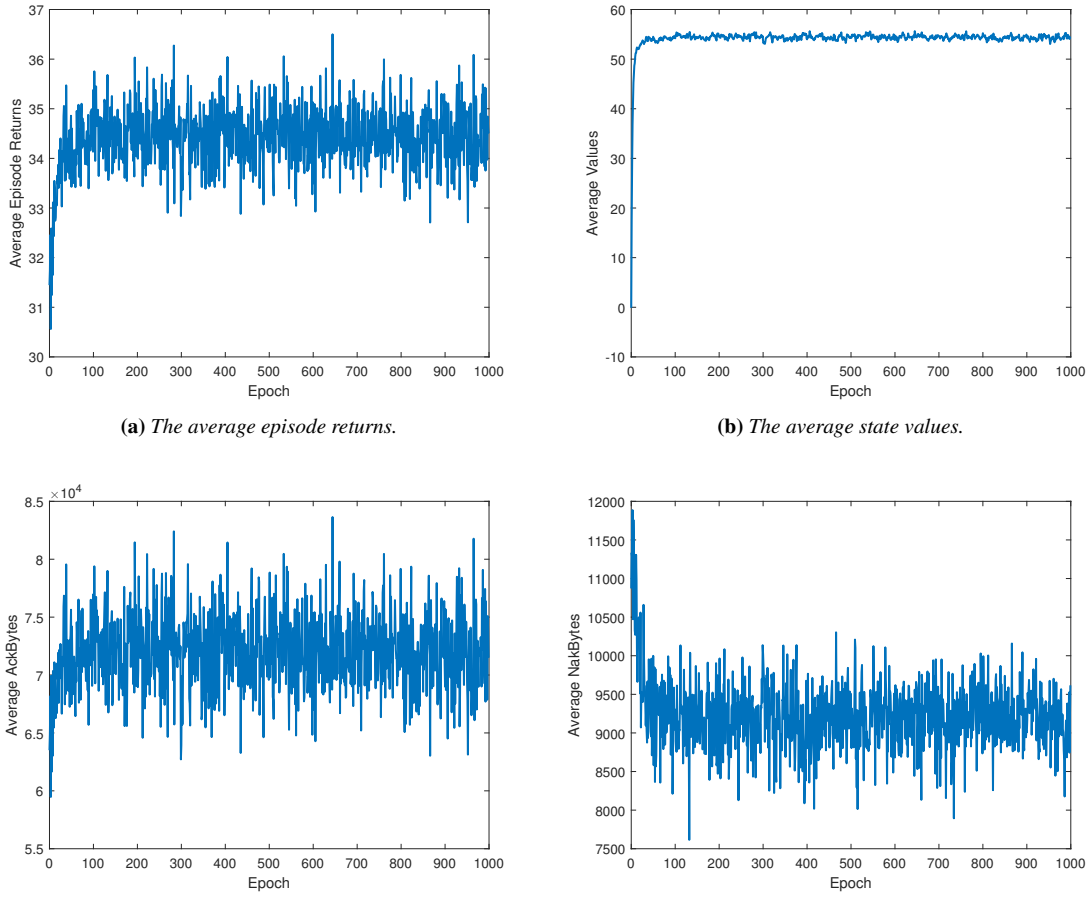
Here, $r_t^n$ is the reward in TTI $t$ for the episode $n$. $\text{AckBytes}_t^q$ and $\text{NakBytes}_t^q$ are the number of acknowledge bytes and negative acknowledge bytes for user $q$, respectively. $N$ is the number of collected episodes.

Figure 7 demonstrates the training performance of TDRL algorithm. The evolution of average episode returns is shown in Figure 7a. Starting from an average of around 30.5, the algorithm converges to a average episode returns of 34.5. Note that the PPO approach updates the network parameters in an offline way, so we can utilize multi-episode data per epoch, which guarantees an unbiased gradient direction. After about 50 epochs, the average returns curve converges. The fluctuation after convergence is to be expected, since the channel condition during each TTI is generated with diversity. Besides, the curve of the average state values converges at the same pace as that of the average returns, as presented in Figure 7b. Compared to Figure 7a, this curve is smoother as state is the only consideration. Figure 7c and Figure 7d show the overall trends of acknowledge bytes and negative acknowledge bytes, respectively. Rising from an average of around $5.9\times10^4$, the algorithm converges to a mean acknowledge bytes of $7.2\times10^4$. Also, the stable value of negative acknowledge bytes is about 9200, a drop from an initial average of 11900. These two curves illustrate that the algorithm is going through trial and error in the early learning stage. Subsequently, the algorithm learns the optimal RA strategy through constant interactions. The optimal RA strategy reduces the overall co-channel interference level, thereby facilitating correct signal transmission and decreasing the retransmission queue.

Figure 8 shows the testing performance of real-time resource allocation based on the TDRL algorithm. Consistent with the training process, we set the maximum steps per epoch to 100, that is, we verify the real-time performance for a continuous 100 milliseconds. After each epoch, we reset the environment to start a new round of testing, for a total of 1000 rounds. Figure 8a and Figure 8b show the average acknowledge and negative acknowledge bytes, from which we



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**(a)** *The average episode returns.*

**(b)** *The average state values.*

**(c)** *The average ackbytes.*

**(d)** *The average nakbytes.*

**Figure 7.** *The training performance of the TDRL algorithm.*

can see that the testing performance shows a stable trend. In Figure 8a, the average acknowledge bytes have a mean of $6.862 \times 10^4$, which is slightly worse than the convergence performance of $7.2 \times 10^4$. The reason for this performance drop is that we cannot guarantee that the testing and training processes are in the exact same communication environment, and the unknown environment makes the policy occasionally perform poorly. Furthermore, RL methods focus on policy learning. Even in the same environment, different RA decisions may be made. The above intrinsic reasons make a little difference in training and testing performance, which we consider to be tolerable. In Figure 8b, the average negative acknowledge bytes are at around 9037, and this is basically stable with the training result of 9200.

To make our proposed algorithm more convincing, we also verify the real-time resource allocation performance under three bandwidths (1.4, 5 and 10 M), and the tested spectrum efficiencies (bps/Hz) are shown in Table 4.

**Table 4.** *Spectrum efficiency with different bandwidths.*

| Bandwidth | 1.4 M | 5 M | 10 M |
|---|---|---|---|
| Spectrum Efficiency | 1.98 | 2.12 | 2.37 |

## 5.3 Algorithm Comparison

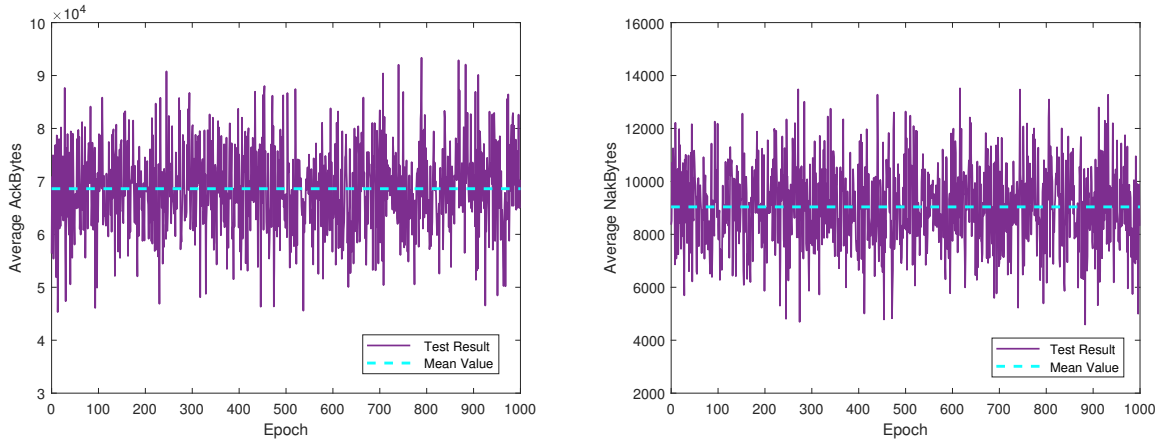
In this subsection, we compare the TDRL algorithm with other algorithms in average episode returns, fairness and complexity.

The algorithms compared with our proposed TDRL algorithm for resource allocation (**TDRL-RA**) are:

1) PPO algorithm for resource allocation (**PPO-RA**): Instead of the Transformer architecture, the actor network is a fully connected network with the same



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**(a)** *The average ackbytes.*

**(b)** *The average nakbytes.*

**Figure 8.** *The testing performance of the TDRL algorithm.*

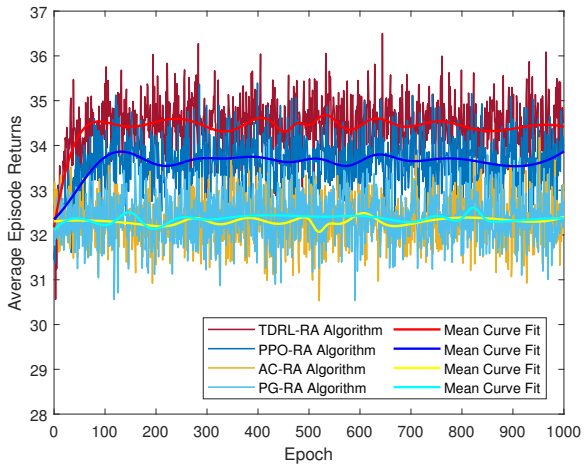



**Figure 9.** *The convergence of different RA algorithms.*

structure as that of the critic network.

2) Actor-critic algorithm for resource allocation (**AC-RA**): In this scheme, the algorithm updates the network parameters online. Hence, data is used only once and then discarded. Apart from this, the restriction of KL is removed in the gradient optimization.

3) Policy gradient algorithm for resource allocation (**PG-RA**): The PG-RA algorithm break the actor-critic framework, and the actor network remains. Also, the algorithm optimizes network parameters online without KL constraints.

• **Average Episode Returns:** Figure 9 illustrates the average episode returns achieved by the above four algorithms. It can be observed from Figure 9 that the convergence performance of the proposed algorithm outperforms the PPO method in terms of convergent speed and convergence value. The Transformer-based PPO scheme converges to a average episode returns of 34.5 after 50 epochs, whereas the PPO algorithm without Transformer achieves a value of 33.5 after 100 epochs. The attention mechanism plays a key role in explaining this result. The attention can avoid invalid action exploration, such as allocating resources for inactive users, thus accelerating the convergence of the algorithm. In addition, the AC and PG algorithms in Figure 9 cannot converge during the training process, proving that it is inefficient to update network parameters online. The average episode returns of AC and PG algorithms constantly fluctuate around the value of 32.3. These approaches with low data utilization cannot cope with the MDP problems with large-scale state and action spaces.

• **Algorithm Complexity:** We also analyze the complexity of the four RA algorithms, as presented in Table 5 where $D$ is the dimension of $x_t^q$. Tricky masking schemes and user profiling technique is able to simplify the MSMU-RA problem by greatly reducing the state and action spaces, which means that our proposed algorithm can make better decisions in the face of complex and changeable cellular communication environments.

Also, we compare with the traditional resource scheduling algorithms (RR [31], Max C/I [32] and PF [33]) in terms of fairness.

• **Fairness Performance:** Table 6 summaries the



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**Table 5.** *Complexity comparison of different algorithms.*

| Algorithm | Complexity |
|---|---|
| PG-RA | $\mathcal{O}\left(\left(U+C+UD\right)\left(BC\left(U/B+1\right)\right)\right)$ |
| AC-RA | $\mathcal{O}\left(\left(U+C+UD\right)\left(BC\left(U/B+1\right)\right)\right)$ |
| PPO-RA | $\mathcal{O}\left(\left(U+C+UD\right)\left(BC\left(U/B+1\right)\right)\right)$ |
| TDRL-RA | $\mathcal{O}\left(\left(U+C+U\right)\left(BC\left(U/B+1\right)\right)\right)$ |

**Table 6.** *Fairness performance of different RA algorithms.*

| Algorithm | RR | PF | Max C/I |
|---|---|---|---|
| Fairness | 0.701 | 0.315 | 0.119 |
| PG | AC | PPO | TDRL |
| 0.483 | 0.595 | 0.703 | 0.708 |

fairness performance of different RA algorithms. It can be clearly observed that the fairness of Max C/I algorithm is the worst with an index of $0.119$, because it only takes the throughput performance into consideration. To be consistent with the above introduction, RR algorithm has the largest fairness index in the traditional methods. Moreover, the fairness performance of data-driven approaches is better than that of PF and Max C/I algorithms, and is comparable to RR algorithm. In particular, our proposed algorithm is the most prominent with a value of $0.708$, which is slightly larger than $0.701$ of RR algorithm. The simulation results demonstrate that the TDRL algorithm show good fairness performance without sacrificing the SE performance.

## 5.4 Parameter Analysis

In this section, we analyze the impact of key parameters on our proposed TDRL algorithm, including the network structure, learning rate, discount factor, episode length and clip ratio.

As depicted in Figure 10a, we compare the average episode returns of the TDRL algorithm under three network architectures. It can be seen that all three network structures have excellent performance. For the feature extraction and processing capability required for state information, we have selected the network structure setting of the blue curve to prevent occasional failure of the TDRL algorithm.

As presented in Figure 10b, we compare the average episode returns of the TDRL algorithm under three learning rate settings. The learning performance is the worst with a convergence value of $37$ under the settings where the learning rates of the actor network and critic network are $3\times10^{-5}$ and $1\times10^{-4}$, respectively. This is because the learning rate is so conservative that the optimal RA policy is not explored. The TDRL algorithm converges fastest when the learning rates of the actor network and critic network are $3\times10^{-3}$ and $1\times10^{-2}$. However, the late learning converges around $38$, which is $0.5$ away from the optimal value. The reason for the performance gap is that the learning rate is too large, which causes the algorithm to oscillate near the optimal solution.

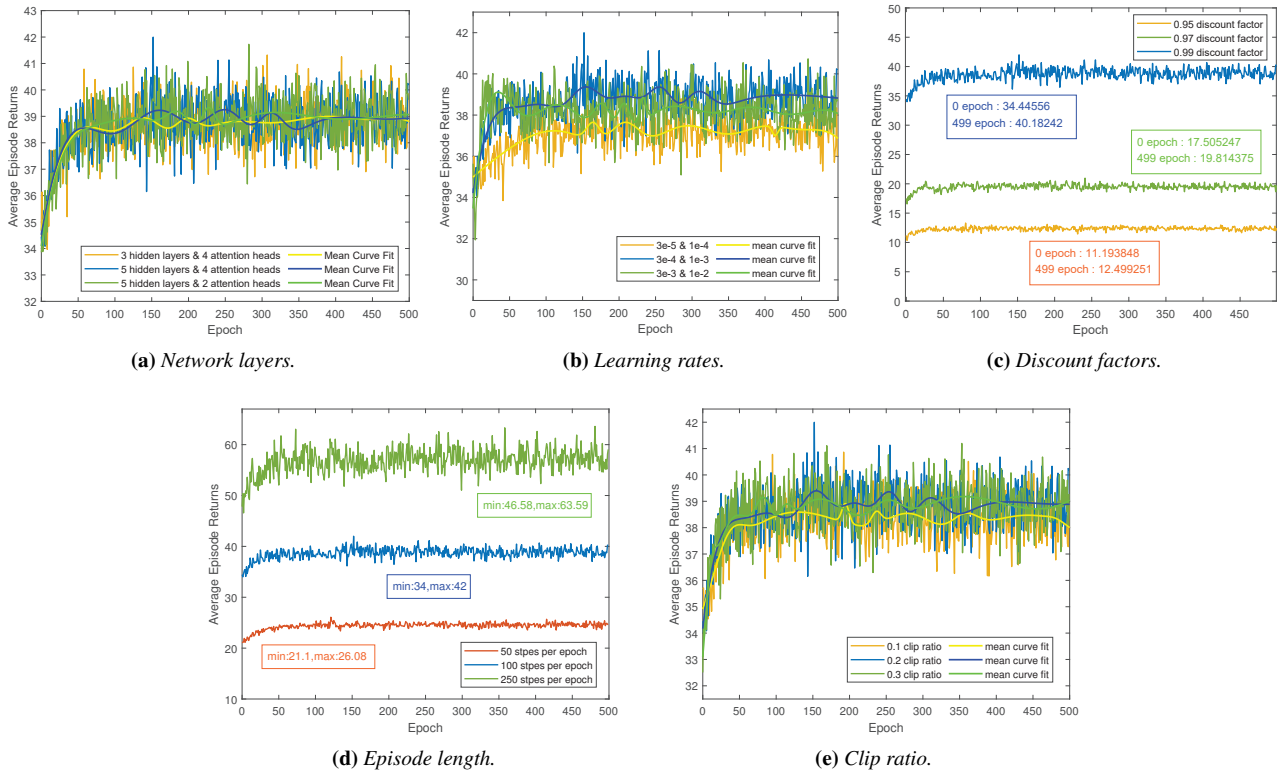
As illustrated in Figure 10c, we compare the average episode returns of the TDRL algorithm under three discount factors. The increase of average episode returns is minimal at the learning rate of $0.95$, from an initial value of $11.1938$ to a convergence value of $12.4993$. It can be seen that the increase in average episode returns will be more clear with the increase of learning rate. The average episode returns ranges from an initial $34.4456$ to $40.1824$ at convergence with a learning rate of $0.99$, which is more desirable. Generally speaking, too large discount factor is easy to lead to the difficulty of convergence. Whereas, too small discount factor drowns the feedback a little farther away. A suitable discount factor is chosen to be as large as possible under the premise that the algorithm can converge.

As shown in Figure 10d, we compare the average episode returns of the TDRL algorithm at three episode lengths. Similar to the discount factor, the episode length also determines the size of the average episode returns. Obviously, the longer the episode, the clearer the increase. Of course, the longer episode is not always better. We observed that the green curve oscillates too strongly at an episode length of $250$, so it is not selected by us. If the episode is too short, the interaction trajectories cannot be easily distinguished. To sum up, we finally set up an epoch with $100$ steps.

In Figure 10e, we compare the average episode returns of the TDRL algorithm with three clip ratios. Compared to clip ratios of $0.2$ and $0.3$, the performance of average episode returns with a clip ratio of $0.1$ is slightly inferior. This is because a too small clip ratio focuses on keeping the training stable and thus lacks sufficient exploration of the environment. Meanwhile, we do not expect excessive data saving to lead to a biased RA policy. Based on the above analysis, we suggest that a clip ratio of $0.2$ is adopted.



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**(a)** *Network layers.* **(b)** *Learning rates.* **(c)** *Discount factors.*

**(d)** *Episode length.* **(e)** *Clip ratio.*

**Figure 10.** *The generalization performance of spectral efficiency.*

## 5.5 Generalization Performance

We compare the TDRL algorithm with other methods in terms of user density, cell coverage radius, traffic intensity and transmit power. Except for the parameter to be compared, others are set according to Table 2.

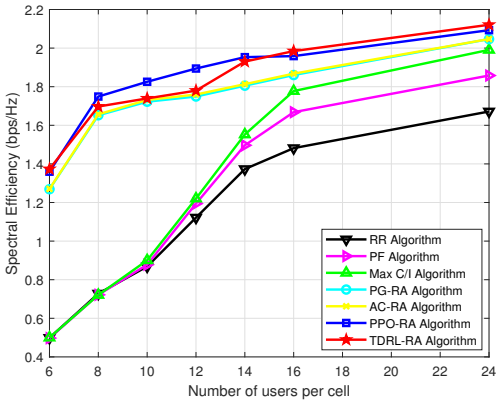
*1) User Numbers:* The number of users per cell has an impact on the user distribution, which is significant for the generalization of the SE performance. As presented in Figure 11a, the number of users per cell ranges from 6 to 24. It can be seen that the system SE shows an increasing trend with the increase of the number of users. Moreover, the performance of the DRL-based schemes obviously exceeds that of the traditional schemes, particularly, the gap can reach 0.6-0.8 bps/Hz in scenarios with low user density. In DRL-based schemes, the PPO-RA approach obtains the best SE performance in $[8, 14]$. Whereas, our proposed algorithm performs well in the case of 16 to 24 users per cell, which proves that the TDRL algorithm maintains a stable and excellent level in large-scale multi-user cellular networks.

*2) Cell Coverage Radius:* As shown in Figure 11b, as the cell coverage radius becomes larger, the SE of the system tends to increase first and then remain stable. Since users are uniformly located in the cell, the increase in the cell coverage radius may cause the inter-cell interference to decrease, thereby improving SE. When the cell coverage radius is 50 m, the Max C/I algorithm achieves the best SE performance because the Max C/I algorithm always schedule users with the best channel quality to ensure the maximum system performance. The advantage of Max C/I algorithm is especially obvious in the case of severe interference, and the advantage gradually weakens as the interference decreases. With the expansion of cell coverage, it is obvious that the SE performance of our proposed algorithm surpasses that of other methods, and even exceeds the traditional schemes by 0.4-0.6 bps/Hz. This proves that the employment of attention can explore scheduled dependencies of resources to make RA decisions that can avoid interference.

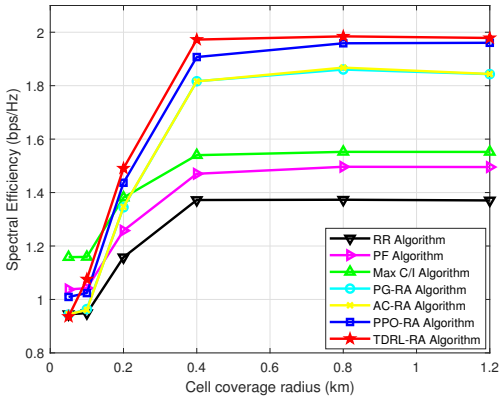
*3) Traffic Intensity:* To be consistent with Section 5.1, Poisson arrival and exponential call-holding time statistics are carried out. In our simulation, the ON/OFF traffic model is adopted. We define traffic



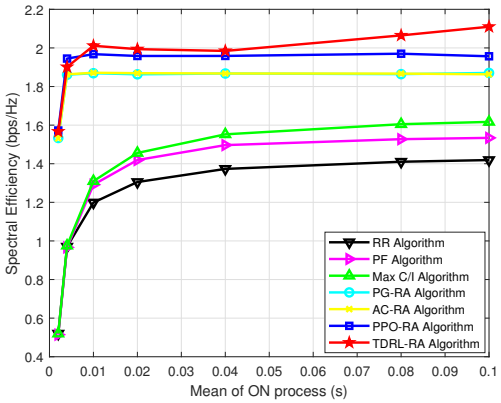
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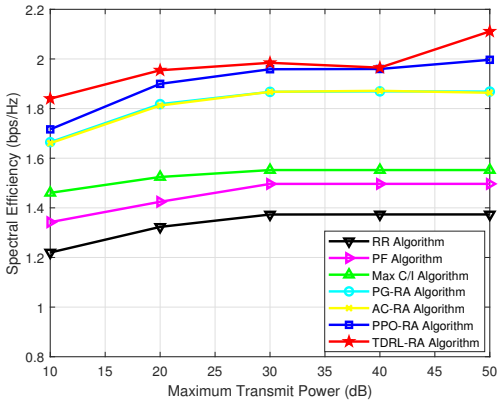

**(a)** *Spectral efficiency with different number of users per cell.*




**(b)** *Spectral efficiency with different cell coverge radius.*




**(c)** *Spectral efficiency with different mean of ON process.*




**(d)** *Spectral efficiency with different maximum transmit power.*

**Figure 11.** *The generalization performance of spectral efficiency.*

intensity as

$$\mathrm{I} = \frac{\mu}{\mu + \zeta}, \tag{33}$$

where $\mu$ is the mean of ON process, and $\zeta$ is the mean of OFF process. Figure 11c illustrates the variation of the SE performance with the mean values of the ON process. Here, $\zeta$ is set to a fixed value of $0.004$ s. When the traffic intensity is $0.5$ with a $0.004$ s ON process mean, the SE performance difference between the traditional schemes and the DRL-based schemes is as much as $1$ bps/Hz indicating that the dynamic RA method has greater allocation flexibility to achieve higher resource utilization. Additionally, the simulation results demonstrate that the TDRL algorithm show good generalization performance on different traffic intensities.

*4) Transmit Power:* To test the generalization of the proposed approach, we also plot the SE achieved across $10000$ TTI for maximum transmit power ranging from $10$ to $50$ dBm. As the transmit power increases, the SE increase slightly and then stabilizes as presented in Figure 11d. As we can see from Figure 11d, all four DRL methods demonstrate excellent performance across the entire range, and the SE of the DRL methods outperform that of the traditional schemes by $0.2$-$0.6$ bps/Hz. In particular, the TDRL algorithm has the highest SE performance, and also shows good generalization performance on different transmit powers.

## VI. CONCLUSION

In this paper, we consider that the MSMU-RA problem in cellular networks is a dual-sequence decision-making process, and thus present the TDRL algorithm to find the optimal RA policy for maximum SE and user fairness. We utilize RL tools to describe the



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dynamic interactions between the agent and the environment, and optimize the RA strategy based on the "observation-actor-critic-returns" framework. Furthermore, we present a Transformer-based actor structure for decision-making, and employ the attention mechanism to explore spatial correlations and scheduled dependencies. Simulation results show that our proposed approach outperforms both the traditional and DRL methods in SE and fairness performance. In addition, the results verify that the TDRL algorithm is superior to DRL approaches in terms of convergence speed and generalization.

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## Biographies



***Zhao Di*** received the B.S. degree on Communication Engineering from Shandong Normal University, Jinan, China, 2018. She is currently working towards her Ph.D degree from Xidian University, Xi'an, China. Her research interests include wireless resource management, satellite communications and deep reinforcement learning in wireless networks.



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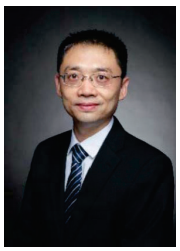
***Zheng Zhong*** received the B.S., and Ph.D degrees in communication and information systems from University of Science and Technology of China, and Peking University, in 2009 and 2018, respectively. In 2018, he joined the Institute of Telecom and Navigation Satellite, China Academy of Space Technology. His research interests include satellite communications and free space optics communications.



***Qin Pengfei*** received the B. Eng., and Ph.D degrees from College of Information Science and Electronic Engineering, Zhejiang University, China in 2014, and 2020, respectively. He was a visiting student in Singapore University of Technology and Design, Singapore from May 2018 for 4 months. His current research interests focus on satellite communications, flexible payload, microwave devices and wireless resource allocation.



***Qin Hao*** received the B.S., M.S., and Ph.D degrees in communication and information systems from Xidian University, Xi'an, China, in 1996, 1999, and 2004, respectively. In 2004, he joined the School of Telecommunications Engineering, Xidian University, where he is currently an Associate Professor of communications and information systems. His research interests include wireless communications and satellite communications.



***Song Bin*** received the B.S., M.S., and Ph.D degrees in communication and information systems from Xidian University, Xi'an, China, in 1996, 1999, and 2002, respectively. He is currently a Professor with Xidian University. He has authored over 100 journal articles or conference papers and holds 50 patents. His research interests include multimedia communication, multimodal data fusion, content-based image recognition and machine learning, reinforcement learning, the Internet of Things, big data, and recommendation systems.



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